TERROCHROMA

Interface & Navigation Design Specification

Soil-Inspired Color System • Native Subscription Model

Document ID	UI-SPEC-TC-2026-V1
App Name	TerroChroma – Rapid Optical Soil Carbon Sensing
Platform	Kotlin Multiplatform (Android + iOS)
Monetization	Native Only (Google Play Billing + StoreKit 2)
Version	1.0 | October 2026
Status	Ready for Implementation

This document defines the complete visual language, navigation structure, screen hierarchy, interaction patterns, and native subscription integration for the TerroChroma mobile application. The design prioritizes outdoor readability, one-handed field use, and a warm soil-and-vegetation color system that reinforces the product's connection to land and regenerative agriculture.

1. Color System – Soil & Vegetation Palette

The entire interface is built on a restrained palette of deep soil browns and living greens. These colors evoke healthy earth and growing vegetation while providing excellent contrast for outdoor sunlight readability.

Color Name	Hex	Role	Usage
Deep Soil	#3E2723	Primary Dark	Headers, strong text, icons
Soil Brown	#5D4037	Primary Brand	App bars, primary buttons, nav
Mid Soil	#6D4C41	Secondary	Secondary text, borders
Leaf Green	#2E7D32	Primary Accent	Success states, active tabs, CTAs
Mid Green	#388E3C	Interactive	Selected items, progress
Olive	#689F38	Highlight	Badges, free-trial tags
Cream	#F5F5F0	Background	Main screen backgrounds
Warm Beige	#EFEBE9	Surface	Cards, bottom sheets

Usage Rules: Primary actions and success states use Leaf Green. Navigation bars and strong structure use Soil Brown. Backgrounds stay in the warm cream/beige family to reduce glare. Never use pure white or pure black as large surfaces.

2. Overall Navigation Structure

Primary navigation uses a persistent bottom bar with four destinations plus a prominent center Floating Action Button (FAB) dedicated to the core Capture action. This keeps the most frequent field workflow one tap away at all times.

Tab	Purpose	Primary Actions
Home	Dashboard, recent samples, quick stats	View last result, start new capture
Capture (Center FAB)	Full-screen viewfinder + guided wizard	Align card → Capture → Infer → Save
Map	Interactive field map of all samples	Filter pins, view details, export selection
Library	Full sample history & bulk management	Search, filter, multi-select export
Settings	Preferences, calibration, account, billing	Manage subscription, models, sync

Navigation Principles: Bottom bar remains visible on all primary screens. Capture opens as a full-screen modal experience. Secondary screens (sample detail, paywall, settings sub-pages) use standard back navigation or bottom sheets.

3. Screen-by-Screen Interface Vision

3.1 Home / Dashboard

Warm cream background. Top app bar in Soil Brown with subtle green accent line. Large “Last Sample” card shows SOC %, SOM %, confidence interval, and a thumbnail of the processed soil region. A horizontal stats row displays samples taken today / this week and average SOC. Below sits a scrollable list of recent sample cards (swipeable). The center FAB (Leaf Green) is always present for new capture.

- Header shows current location context + offline/online indicator
- Last Sample card uses soft elevation and Warm Beige surface
- Empty state gently prompts the user to take their first sample

3.2 Capture – Guided Viewfinder (Core Experience)

Full-screen camera viewfinder with live overlays. The design is deliberately minimal so the soil and calibration card remain the visual focus.

- Real-time alignment guides highlight ArUco / AprilTag corners on the calibration card
- Digital spirit-level bubble (IMU-driven) turns green when pitch/roll ≤ 10°
- Ambient light quality meter warns if lighting is too harsh, dim, or uneven
- Status chips: “Level OK”, “Light OK”, “Card Detected” appear only when relevant
- Large circular shutter button (Leaf Green) becomes active only when all gates pass
- Capture sequence: Card Alignment → Stability Check → RAW Lock → Local Inference → Result

3.3 Result Overlay / Bottom Sheet

After inference completes (target < 2 s), a bottom sheet rises with generous spacing:

- Large SOC % number (primary) and SOM % (secondary)
- Carbon stock (Mg C/ha) when bulk density is available
- Confidence / uncertainty band displayed as a clear visual range
- Thumbnail of the segmented soil ROI + key color metrics (L*, Munsell Value)
- Primary actions: Save & Tag • Retake • Add Note / Moisture
- Secondary: Share / Export this sample

3.4 Map

Full interactive map using platform map SDKs. Sample pins are color-coded by SOC range using the green–brown scale (low carbon = lighter soil brown, high carbon = deeper leaf green). Tapping a pin opens a compact bottom sheet with summary metrics and a jump-to-detail action. Filters (date, SOC threshold, sync status) sit in a collapsible top bar.

3.5 Library / History

Searchable and filterable list (or optional grid) of all samples. Each card shows thumbnail, date/location, SOC value, and sync status badge. Multi-select mode enables bulk export (GeoJSON / KML / CSV) or deletion. Detail view expands to full metrics, feature vector summary, provenance hash, map snippet, notes, and image gallery.

3.6 Settings

Grouped into clear sections: Account & Subscription, Calibration Cards, Models & Regional Overlays, Units & Display, Offline Sync & Storage, Hardware Diagnostics (camera lock test, IMU, BLE spectrometer pairing), and About / Legal. Subscription management deep-links to the native store subscription screen.

4. Detailed Capture & Processing Flow

Step	User Sees	System Behavior
1. Open Capture	Full-screen viewfinder + overlays	Lock camera parameters (ISO, shutter, WB). Start IMU & light monitoring.
2. Align Card	Corner guides + “Move card into frame”	Detect ArUco/AprilTags. Perspective correction preview.
3. Stability Gate	Spirit level bubble + status chips	Block shutter if tilt > 10° or light unstable.
4. Capture	Shutter animation + brief freeze	Capture RAW (DNG/Bayer) + JPEG proxy. Tag GNSS + IMU metadata.
5. Processing	Progress ring + “Analyzing soil…”	Color calibration → L*a*b*/GLCM → EPO moisture correction → on-device ML inference.
6. Result	Bottom sheet with SOC / SOM / confidence	Display results. Offer Save, Retake, or Add metadata.
7. Save	Confirmation toast + return to Home/Map	Encrypt & store locally. Queue for cloud sync if online.

5. Native-Only Subscription Model

Subscriptions are implemented exclusively with platform-native billing systems: **Google Play Billing Library** on Android and **StoreKit 2** on iOS. No third-party subscription management platforms are used.

5.1 Recommended Plan Structure

Plan	Suggested Price	Unlocks
Free	$0	Limited captures per month + basic results
Pro Monthly	$9.99 – $14.99	Unlimited captures, full history, map, exports, priority models
Pro Yearly	$79 – $99 (best value)	Same as Monthly + discount. Optional free trial.

5.2 Price & Offer Management

All prices, free trials, introductory offers, and promotional codes are managed directly inside Google Play Console and App Store Connect. Changing a price or trial length does not require an app update; the stores push the new commercial terms to the billing libraries. The app only needs to query the current product details at runtime.

5.3 In-App Experience

- Soft paywall appears after the user has taken a small number of free samples (value-first)
- Paywall screen uses the same soil-green visual language; clear yearly savings callout
- “Restore Purchases” button is always visible and easy to find
- Active subscription status is checked on launch and before gated actions
- “Manage Subscription” deep-links to the native store subscription management UI
- Entitlement checks are performed locally after the billing library confirms status

5.4 Technical Notes (Native Only)

Android: Use Google Play Billing Library 7+ with a clear ProductDetails → purchase flow and acknowledge/consume handling. iOS: Use StoreKit 2 (Product, Transaction, RenewalInfo) with StoreKit Testing in Xcode for development. Both platforms require server-side receipt validation for production security; a lightweight backend endpoint is still recommended even with a pure native client approach.

6. Core Design Principles

• **Outdoor First** — High contrast, large type, no large pure-white surfaces. Status colors remain distinguishable under bright sun.

• **One Primary Action** — Every screen has one obvious next step. Capture is always one tap away via the center FAB.

• **Error Prevention** — Capture is blocked until tilt, light, and card detection are satisfied. Messages are short and non-technical.

• **Offline by Default** — All core flows work without network. Sync status is visible but never blocking.

• **Progressive Disclosure** — Field technicians see a simple path. Agronomists can expand to raw L*a*b*, GLCM, model ID, and provenance.

• **Warm & Grounded** — Soil browns and living greens reinforce the product's connection to land. Avoid cold greys and pure blacks.

• **Thumb-Friendly** — Critical controls sit in the lower half of the screen for comfortable one-handed use in the field.

7. Document Control

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Related Docs	SRS-SOC-2026-V1 • Technical Blueprint
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End of Document — TerroChroma Interface & Navigation Design Specification